

A Smart Multi-User Wireless Nurse Calling System and E-notice Board for Health Care Management

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Abstract. To provide smart healthcare service in a hospital, a prototype of fingerprint recognition based multi-user wireless nurse calling system has been designed and presented in this paper. The proposed system also has an attractive feature to circulate any common information or emergency message to the patients. This message can be sent by any authorized doctor or hospital management authority from any place within the LAN network. Use of fingerprint makes the system more robust and secure i.e. there is no chance to access the system by any unauthorized person. The health care management system can work up to 700 meters circular distance. The Arduino, ATmega328P, nRF24L01+PA+LNA, fingerprint scanner, LED display, speaker etc. hardware components have been used to build the prototype, these hardware accessories are commercially available to anywhere and comparatively low cost. Instead of conventional wired setup, wireless healthcare management system ensures easy maintenance, simpler and fast deployment. Flexibility and versatility is guaranteed by wireless tuning. Since the proposed system prototype shows satisfactory performance during practical trials, the system can be considered as a suitable candidate for the health care management system of a hospital or any healthcare center.

Keywords: Fingerprint Recognition, Calling System, Electronic Notice Board, Microcontroller, WiFi.

1 Introduction

In the present era of technological revolution, folk are searching for such a smart electronic system which can provide assurance of easy and smooth services with proper authenticity [1][2]. In this article, we are talking about a fingerprint based multi-terminal wireless nurse calling system with electronic notice board which is designed to assist caregivers. A nurse calling service is simply a technical solution for serving the needs of individuals during treatment in a hospital or medical care center. A hospital authority chooses the appropriate system according to their needs. Wireless nurse calling systems are scalable from a few beds to so many [3]. Conventional nurse calling systems use push button switches placed at the patient's room. When it is pressed by any patient, it alerts the central nurse control room [4]. However, in the conventional system, anyone can use it and may deliberately create chaos, leaving a security threat and improper use may result in haphazard use at the nurse station [5]. To mitigate such problems, there is a need to develop an authenticity verification feature in the system where only certain authorized users can access it. Biometrics is behavioral and physical human properties to be used to distinguish a person to grant

access to a system [6]. A fingerprint scanner is a type of electronic component that identifies and authenticates the fingerprints of an individual to grant or deny access to a system [7]. Some monitoring systems are built on computer networks where there is a necessity of an internet connection. However, our proposed calling system does not depend on internet access. For an internet access dependent system, if internet connection is interrupted, the whole system would be useless whereas our suggested model can still provide nurse calling service smoothly. In most traditional wired systems, the calling button is fixed at a position, so the patient must reach to dial the nurse; this challenge can be overcome by deploying the proposed wireless system. Each calling key in this design is compact and battery driven. As a result, if the electricity goes out, the device will operate on battery power. Whenever a person interacts with the system, the central nurse station is alerted by playing an audible sound and displaying the individual's name (patient or duty doctor), room number. There is also a provision for a real time half-duplex voice communication system and this feature can access duty doctors only and not the patient. There is another part, electronic notice board, which is integrated with the calling system. This electronic notice board can be placed anywhere within the hospital, which is useful to display common messages or emergency notification. However, changing the message every day via analog procedure is a very tedious task, instead using NodeMCU, the authority can change the message using the web browser remotely [8][9]. The E-notice board can be used in the patient room as a medicine reminder or waiting room/corridor of the hospital to get the attention of visitors. The proposed system, however, contributes to reduce the transit time for doctors and nurses by providing the precise details of the patient who pressed the button that appears in the base station with an LED display module. During call setup, the same detailed information is also declared in the speaker. These features emphasize the key factors in emergency situations.

2 Concept Making and System Modeling

In today's rapidly developing technological environment, many smart systems are developed depending on the user's needs [10] [11]. Some are home security, some are IoT devices, some are plant monitoring, etc. In this work, we are intended to develop a fingerprint based multi-terminal wireless nurse calling system and e-notice board for health care management of a hospital. The concept of the healthcare management system is depicted in Fig.1 as a block diagram. This system encourages patients and doctors to use them in a simple and convenient manner, seeking emergency help as soon as possible. With the use of a fingerprint module, it also ensures security issues because biometrics are behavioral and physical human properties to be used to distinguish a person to grant access to systems. Instead of a wired network, the proposed system relies on wireless connectivity. Thus, the system's cost is minimized and the complexities of cables are eliminated. The terminal parts of the wireless calling system are placed in every room in the hospital, which are used by both authorized duty doctors and the patient(s) of that room. First, fingerprints of duty

doctors and admitted patients who wish to call for assistance, are collected by the hospital management authority during the admission of a patient and stored in the system's database. After admission, when a patient calls for assistance the module checks the input fingerprint within the database stored in the system and also categorizes the person between any one of two categories indicated as 'Doctors' and 'Patient'. So when a doctor needs his assistant or a patient needs some help then just scan his/her finger then instantly the name of the doctor/patient and corresponding room number will be displayed to the control room and parallelly a call ringtone will be rung on the speaker. Therefore, the nurses from the control room can easily be notified about the location of the calling person. Moreover, sometimes a doctor may need to talk with central nurse control directly then it can be possible by activating a wireless transceiver module on this device i.e. this device has a feature of two-way communication between doctor and assistant. The electronic notice board is another interesting feature of the proposed system. The E-notice board can control any authorized doctor/nurse from anywhere within the LAN network.

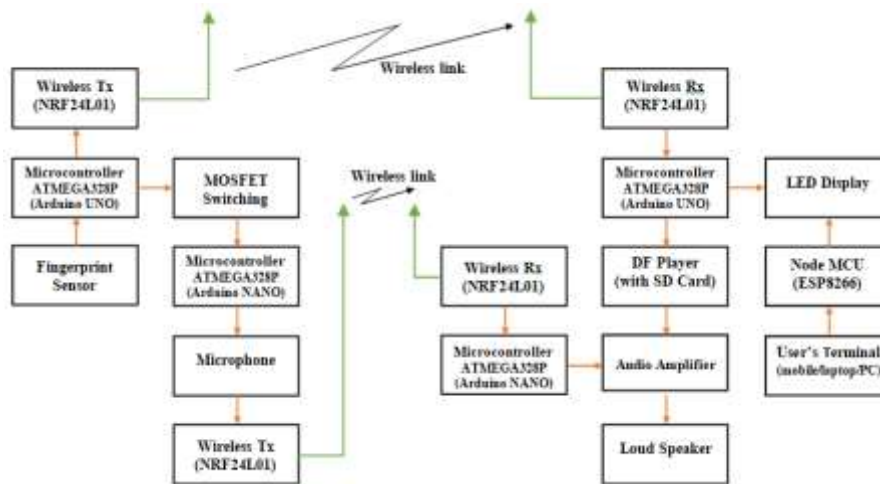


Fig.1. Block diagram of the proposed calling system and e-notice board

In the proposed system, nRF24L01 wireless transceiver module has been used which is controlled by an ATmega328P arduino microcontroller. The fingerprint scanner collects the fingerprint of the calling person, verifies it and communicates with the microcontroller via the universal asynchronous receiver transmitter module (nRF24L01). After receiving an authentication signal from the fingerprint module, the microcontroller determines whether or not to send a voice transfer as well as a command. The transmitting module (via nRF24L01) transmits that command referring to a specific fingerprint through the processing microcontroller (Arduino).

The receiver part will receive the corresponding fingerprint signal via the receiving unit of nRF24L01 (Rx) and process through ATmega328P Arduino microcontroller. According to transmitting corresponding fingerprint commands, it starts to process in the receiving section that was programmed previously. As per the signal, the microcontroller shows a preprogrammed text message on the LED screen and also plays a recorded ringtone stored in the SD card. Simultaneously, if a user needs to display real time, the user must first log in to the particular URL, fill the authentication process and submit the text message. He can also modify the text anytime via a web browser.

3 Accessories of the Proposed System

In this section, the major components like Arduino Uno, fingerprint sensor and radio transceiver module of the designed prototype have been discussed briefly.

3.1 Arduino Uno

The Arduino Uno board, which is a board based on the ATmega328P microcontroller, serves as the system's brain. The ATmega328P is an 8-bit microcontroller operating at 16 MHz [12]. Arduino Uno has only one SPI, one I2C and several GPIO. The serial peripheral interface (SPI) is a synchronous serial communication interface specification used for short distance communication, primarily in embedded systems. Using an SPI link, one master device (typically a microcontroller) controls the peripheral devices [13]. The board includes 32 KB of flash ROM, 2 KB of SRAM, and 1 KB of EEPROM. The Arduino board has a programming environment that allows C or C++ programming language.

3.2 Fingerprint Sensor Module

In any system fingerprint authentication is the act of verifying an individual's identity. In our system, a fingerprint sensor module wired to the arduino, enrolls a new fingerprint and stores directly within the flash memory with different IDs. After uploading the enrolment code, the serial monitor would prompt to insert credentials (fingerprint) to register. This module includes flash memory for storing fingerprints and can be used with any microcontroller or device that supports TTL serial [14]. It is able to store up to 127 different fingerprints. This module communicates with the microcontroller through software/hardware serial. Here, we used software serial by wiring the Tx pin of the fingerprint module to the digital pin-2 of the arduino and the Rx pin of the fingerprint module to the digital pin-3 of the arduino.

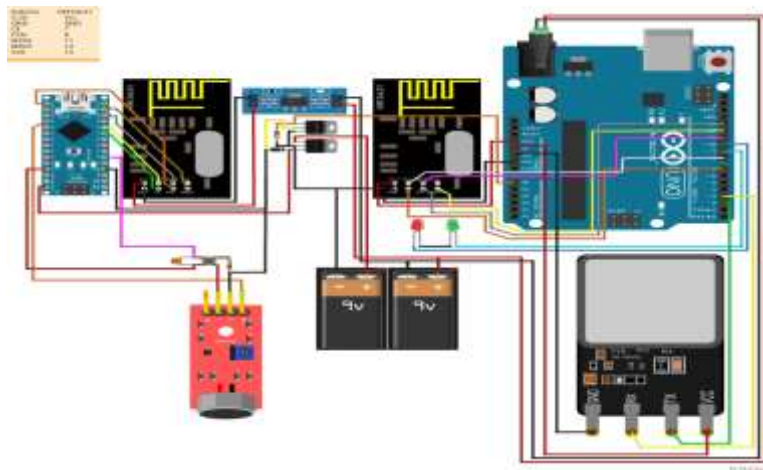
3.3 Radio Transceiver Module

The nRF24L01+PA+LNA module is a radio transceiver module capable of communicating with one or more microcontrollers i.e. it works as a multiceiver. The

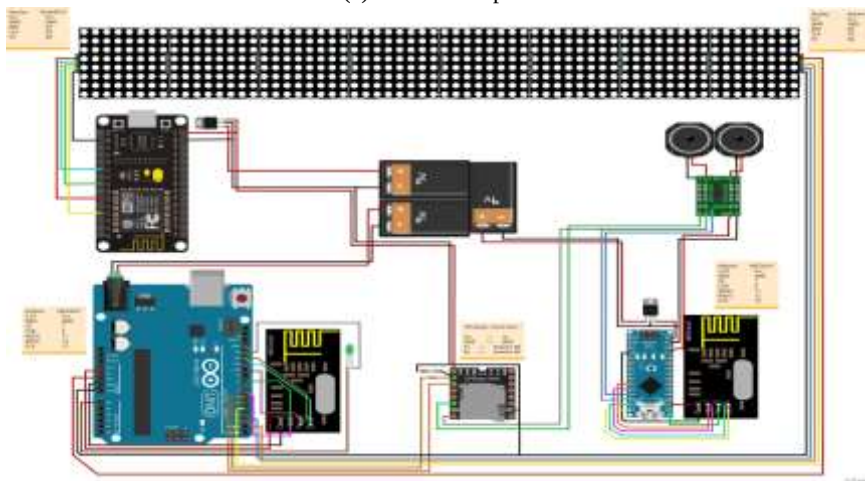
PA means power amplifier which enhances the signal power transmitted from the nRF24L01+. The purpose of the low noise amplifier (LNA) is to receive incredibly weak signals from the antenna and extend it to a better standard. The radio transceiver's talk over distance is approximately 700 meters; this distance can vary depending on the location and atmosphere. Any obstacles can limit the range of the transmitter and receiver. This gives us almost 125 channels with a 1 MHz spacing [16]. The nRF24L01+ transceiver module communicates over a 4-pin SPI with a maximum data rate of 10 Mbps. The channel fulfills a bandwidth of less than 1 MHz at 250 kbps and 1 Mbps air data rate. However, at 2 Mbps air data rate, 2 MHz bandwidth is used. It runs at 250 KHz, 1 MHz, and 2 MHz with a maximum bit rate of 2 Mbps and a transmit frequency of 1 mW in the 2.4 GHz unlicensed band using GFSK modulation. The MISO, MOSI, SCK, CLK pins are used for communication. The NRF CE, SCN pins of nRF24L01 connected to digital pins 7, 8 respectively. The nRF24L01 module uses an IRQ pin for interrupt but it is not used in this scenario as the transmitter module does not need any power saving option. The nRF24L01 module needs 1.9 - 3.6 volts to run. The module has programmable output capacity, which can be set to 0 dBm, -6 dBm, -12 dBm, or -18 dBm, and consumes just 12 mA during transmission at 0 dBm, which is less than a single LED [17]. The Arduino board includes a 3.3V voltage regulator that can control an nRF24L01 module.

4 Project Design and Working Procedure

A schematic circuit diagram of the proposed system has been designed by using "Fritzing" software as shown in Fig. 2 which possesses two major parts: one is master presented in Fig. 2(a) and another is slave/child part depicted in Fig. 2(b). The master part placed at patient's room acts as the main system which can be used by doctors/patients while the slave part placed at the central nurse control station. The master part consists of the following components: a microcontroller (Arduino UNO) linked to the wireless module (NRF24L01), with two LEDs (green and red) that which are used to indicate the user of calling system (patient/doctor) is either authorized or not. Glowing the red LED suggests that the user's fingerprint is not in the database i.e. he/she is not authorized to access the system, while a green LED indicates that the user is an authorized person. The prototype also has a voice communication feature that can use only duty doctors by again providing their fingerprint. When the patient/doctors pressed the fingerprint module, the controller located at the patient's room communicates using the wireless module nRF24L01 to the slave part that is located in the central nurse control room. The slave acknowledges unique addresses of master and opens a channel to establish communication between them. According to unique addresses sent by the master part, the LED screen displays doctor or patient's name with room number. There is no need to call a nurse on foot. By using such wireless electronic displays a nurse from the control station can easily notify the patient about their medicine time if such a display is placed in the patient room.



(a) Transmitter part



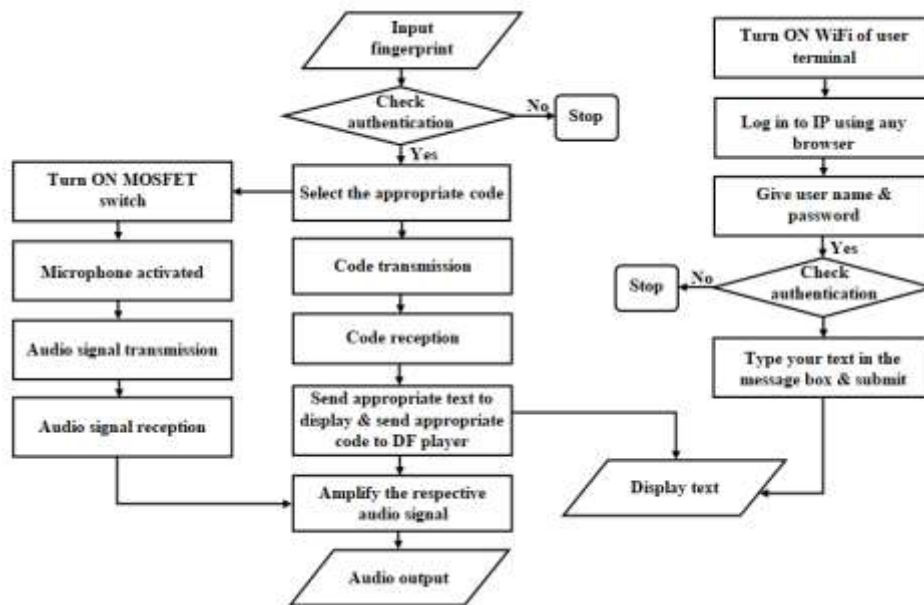
(b) Receiver part

Fig.2. Circuit diagram: (a) Transmitter part (b) Receiver part

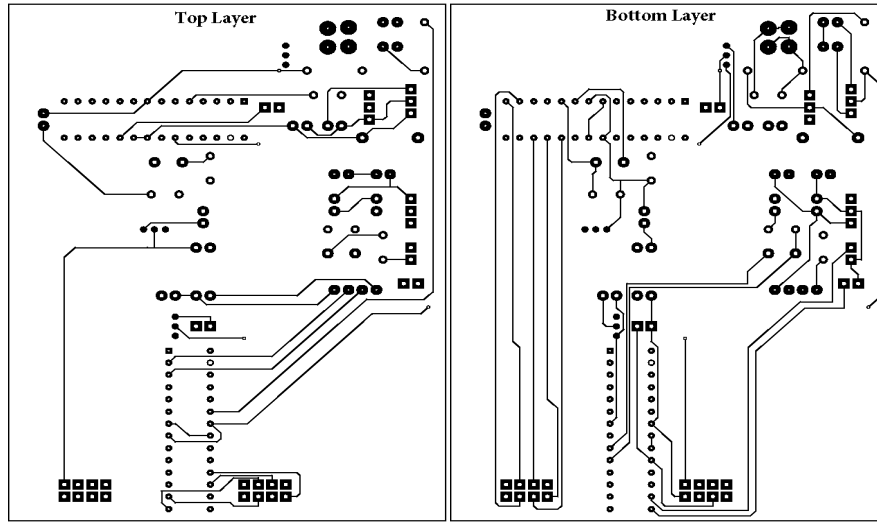
(a) Sign in with user name and password



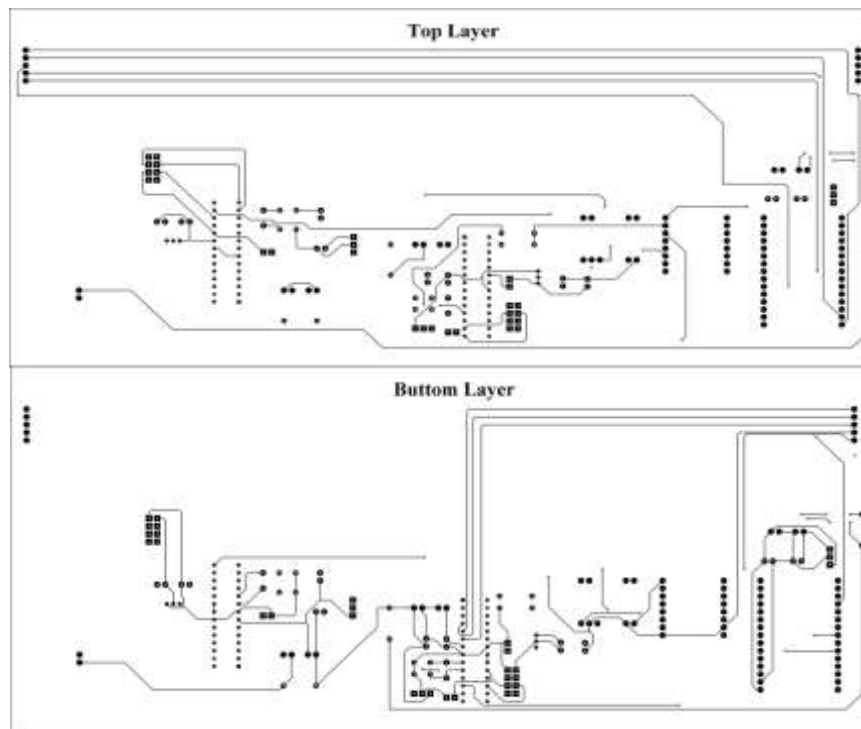
(b) Write notice and submit

Fig.3. Circuit diagram: (a) Sign in with user name and password (b) Write notice and submit**Fig.4.** Flow chart of the of the proposed calling system and e-notice board

Moreover, a centrally common e-notice board can be controlled by the hospital authority to announce important updates for their patient or visitors just by using a web browser for example: “Please leave the patient's relatives before 9 pm” as shown in Fig. 3. The entire signal flow and decision making scenario of the proposed system has been depicted as a flow chart in Fig. 4. The PCB layout and a designed prototype of the proposed calling system and e-notice board have been presented in Fig. 5 and Fig. 6 respectively.

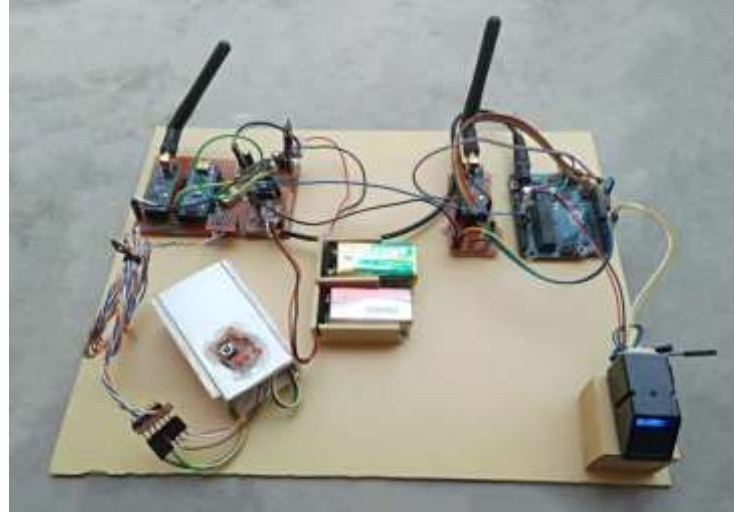


(a) Transmitter part



(b) Receiver part

Fig.5. PCB layout: (a) Transmitter part (b) Receiver part



(a) Transmitter part



(b) Receiver part

Fig.6. Prototype of the proposed system: (a) Transmitter part (b) Receiver part

5 Conclusion and Discussion

A prototype of fingerprint recognition based multi-terminal multi-user wireless smart calling system and electronic notice board for health care management of a

hospital has been developed and proposed in this paper. The system has been implemented by using open source software tools Arduino IDE with C programming language, also used hardware interfaces ATmega328P, nRF24L01+PA+LNA, fingerprint scanner, LED display, speaker etc. For a hospital, a wireless healthcare calling system can play a vital role to provide smart management and service. The designed prototype is secure, stable and economical. It can be tailored to meet specific desires and expectations and suited to any hospital budget. The outcome of the designed system's prototype is impressive during trial. The proposed system ensures confidentiality and reduces human effort in health care system management. This system is not suitable for critical patients because critically injured patients may not be able to use fingerprint scanners.

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